Venkata Lakshmi Parimala Pasupuleti

AI Solutions Engineer · Document Intelligence · Agentic Workflows · LLM API Integration · Python · RAG · Azure · Regulated Domains

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SUMMARY

Senior Software Engineer specializing in production AI systems for regulated, document-heavy environments. Built **Leasel**, an agentic document intelligence platform for compliance operations: OCR-fallback ingestion, section-aware chunking, hybrid search (TF-IDF + semantic reranking), citation grounding to exact source clauses, missing/conflicting clause detection, and an 8-agent LangGraph investigation pipeline - all in an environment where incorrect document interpretation carries regulatory consequences. Built **AgentProbe**, an open-source LLM evaluation toolkit integrating Anthropic, Azure OpenAI, OpenAI, Groq, and Vertex AI with structured output validation and CI regression harnesses. At Entrata (8 years, 100K+ users), owned Python/PHP backend services, async data processing, and regulated data pipelines in a compliance-heavy SaaS environment - 7 years of PHI-encrypted pipelines with zero security incidents. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Document Intelligence · Agentic Document Workflows · RAG · Hybrid Search · OCR · Chunking · Citation Grounding · LLM API Integration · Anthropic · Azure OpenAI · Vertex AI · Python · FastAPI · Django · LangGraph · LangChain · Structured Outputs · Tool Use · Streaming · Async Processing · PostgreSQL · SQLite · Azure · AKS · Service Bus · Azure Document Intelligence · Docker · Kubernetes · CI/CD · GitHub Actions · Terraform · Compliance & Regulated Domains · PHI · HIPAA · SOC 2

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data analysis pipelines, and Python-based API integration layers connecting AI components to application-facing backend services and reporting systems.
- Applying troubleshooting practices across the full stack to improve reliability of data-driven and AI-powered features in production.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built an agentic mission-planning system (LangGraph StateGraph) with **RAG over regulatory documents** - answered operator queries with cited sources from FAA Part 107 corpus; structured outputs enforcing hard constraint validation before tool-invocation and export.
- Deployed an **MCP server** connecting agentic pipeline outputs to downstream enterprise tooling with authentication, execution, failure routing, and observability - production integration following defined API contracts and retry semantics.
- Engineered full document ingestion, chunking, and retrieval pipeline (corpus → FastAPI serving endpoint) with streaming interface for real-time operator responses.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ users, multi-tenant)

- **Owned compliance data pipelines for 7 years** in a regulated environment - HUD income-limit import (few thousand federal rows → ~180,000 domain records), eligibility calculations, and rent workflows powering 100K+ housing units; zero compliance violations across all annual cycles.
- **Built and operated async distributed processing** (RabbitMQ, similar to Azure Service Bus) for tenant record updates - retry logic, idempotency guarantees, and production incident ownership through annual rent-schedule changes.
- **PHI compliance across all resident data**: Libsodium-encrypted pipelines, 100% coverage, zero security incidents; AWS S3 document migration with full metadata validation and customer-facing workflow verification post-cutover.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis and indexing across complex multi-tenant PostgreSQL schemas; shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints.
- Led microservices redesign (Docker/Kubernetes) - 50% scalability improvement, 30% deployment time reduction; CI/CD pipelines and GitHub Actions automation; mentored 5+ engineers across 300+ peer code reviews.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built backend modules (Python, PHP, Java, PostgreSQL) including income-limits import (1.6M HUD/Tax Credit records), document management, and compliance workflows; led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - Agentic Document Intelligence Platform

2026 | Python, LangGraph (8-agent), FastAPI, OCR, SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Document Intelligence surface**: OCR-fallback ingestion (pdfplumber + pytesseract), section-aware chunking, hybrid search (TF-IDF + semantic reranking), **citation grounding to exact source clauses**, missing/conflicting clause detection, and a human-review gate on every regulated decision - 100% citation correctness, 0% unsupported-claim rate on evaluation set.
- **Agentic Investigation surface**: 8-agent LangGraph pipeline with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) producing evidence-backed root-cause hypotheses with HITL escalation when confidence is insufficient.

- Core design rule enforced in code: LLM extracts and reasons; a human or deterministic logic makes every final compliance call. 93.6% precision and 87.9% F1 across 1,240 residential lease PDFs (10 rules, 5 US states + federal).

AgentProbe - Open-Source LLM Evaluation Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- LLM API integration across **Anthropic, Azure OpenAI, OpenAI, Groq, and Vertex AI** - structured outputs, tool use, streaming, model versioning, and cost controls; 15-endpoint FastAPI backend with rate-limit handling and retry semantics.
- 5-metric eval engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression harness with golden datasets and LLM-as-judge scoring for document processing quality verification.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017